

ASTA 607: COMPUTATIONAL STATISTIC 1

QUESTION 1

a. Description

?crabs

b. The crabs data frame has 200 rows and 8 columns, describing 5 morphological measurements on 50 crabs each of two colour forms and both sexes, of the species *Leptograpsus variegatus* collected at Fremantle, W. Australia.

b. Loading the dataset and display of structure

```
data("crabs")
str(crabs)
summary(crabs)
head(crabs)
```

```
data.frame':    200 obs. of  8 variables:
 $ sp      : Factor w/ 2 levels "B","O": 1 1 1 1 1 1 1 1 1 1 1
 ...
 $ sex     : Factor w/ 2 levels "F","M": 2 2 2 2 2 2 2 2 2 2 2
 ...
 $ index: int   1 2 3 4 5 6 7 8 9 10 ...
 $ FL      : num  8.1 8.8 9.2 9.6 9.8 10.8 11.1 11.6 11.8 11.8
 ...
 $ RW      : num   6.7 7.7 7.8 7.9 8 9 9.9 9.1 9.6 10.5 ...
 $ CL      : num  16.1 18.1 19 20.1 20.3 23 23.8 24.5 24.2 25.2
 ...
 $ CW      : num   19 20.8 22.4 23.1 23 26.5 27.1 28.4 27.8 29.3
 ...
 $ BD      : num   7 7.4 7.7 8.2 8.2 9.8 9.8 10.4 9.7 10.3 ...

[1] 200    8
```

sp	sex	index	FL	BD
RW		CL	CW	
B:100	F:100	Min. : 1.0	Min. : 7.20	Min. :
: 6.50	Min. :	:14.70	Min. :	
6.10				
O:100	M:100	1st Qu.:13.0	1st Qu.:12.90	1st
Qu.:11.00	1st Qu.:27.27	1st Qu.:31.50	1st	
Qu.:11.40				
		Median :25.5	Median :15.55	
Median :12.80		Median :32.10	Median :36.80	
Median :13.90				
		Mean :25.5	Mean :15.58	Mean
:12.74	Mean :	:32.11	:36.41	Mean
:14.03				
		3rd Qu.:38.0	3rd Qu.:18.05	3rd
Qu.:14.30	3rd Qu.:37.23	3rd Qu.:42.00	3rd	
Qu.:16.60				

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```
Max. :50.0 Max. :23.10 Max.
:20.20 Max. :47.60 Max. :54.60 Max.
:21.60
```

c. Frequency table

```
table(crabs$sp)
table(crabs$sex)
```

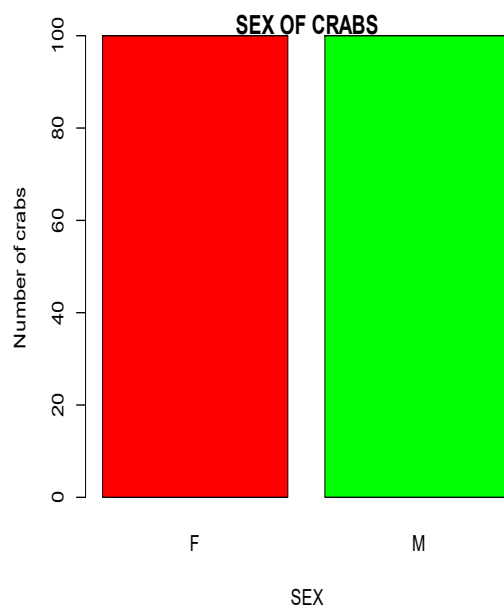
```
  F  M
100 100
```

```
  B  O
100 100
```

d. Bar chart

```
Frequency <- table(crabs$sex)
e. Frequencyy <- table(crabs$sp)

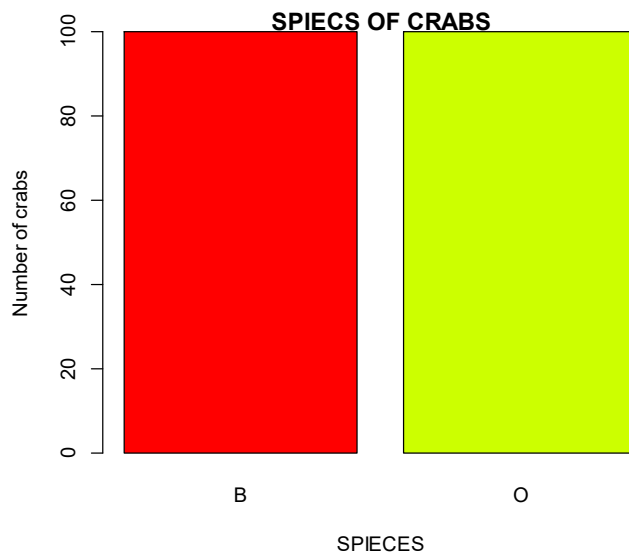
par(mfrow=c(1,1))
barplot(height = Frequency,
        main = "SEX OF CRABS",
        xlab = "SEX",
        ylab = "Number of crabs",
        col = rainbow(3),
        )
```



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```
par(mfrow=c(1,1))
barplot(height = Frequencyy,
        main = "SPIECS OF CRABS",
        xlab = "SPIECES",
        ylab = "Number of crabs",
        col = rainbow(5),)
```

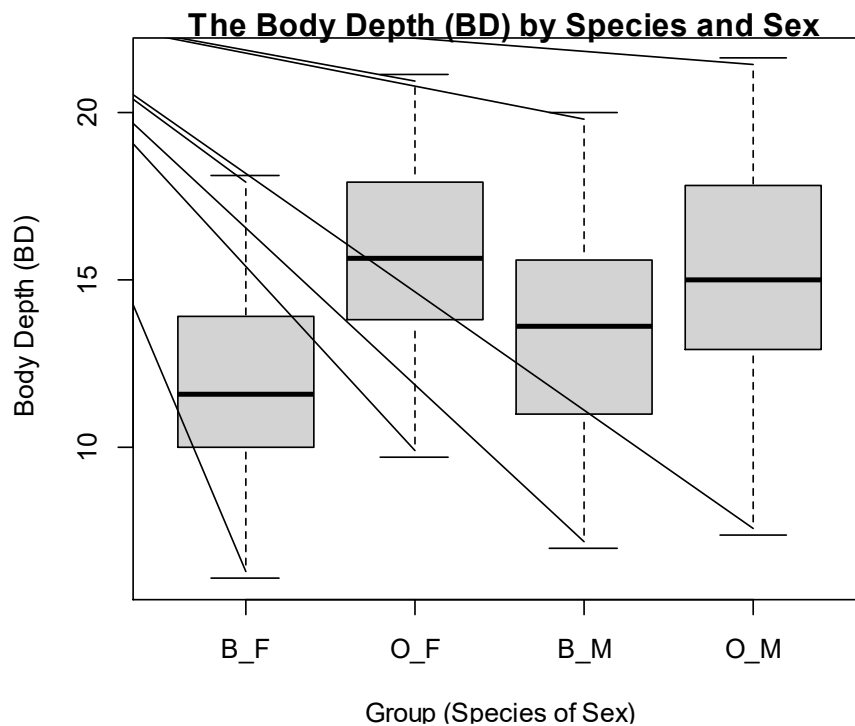


e. The dataset is perfectly balanced — each of the 4 groups has exactly 100 observations, giving an equal. Because the groups are equal in size, any group comparisons (means, spreads, etc.) will not be biased by unequal sample sizes. Statistical tests will have equal power across all 4 groups.

QUESTION 2

a. (a) Side-by-side boxplot of BD for the 4 groups (species x sex)

```
boxplot(BD ~ interaction(sp, sex, sep = "_"), data =
crabs,
        xlab = "Group (Species of Sex)", ylab = "Body
Depth (BD) ",
        main = " The Body Depth (BD) by Species and Sex")
```



b. Identification of outliers and comparison of central tendency

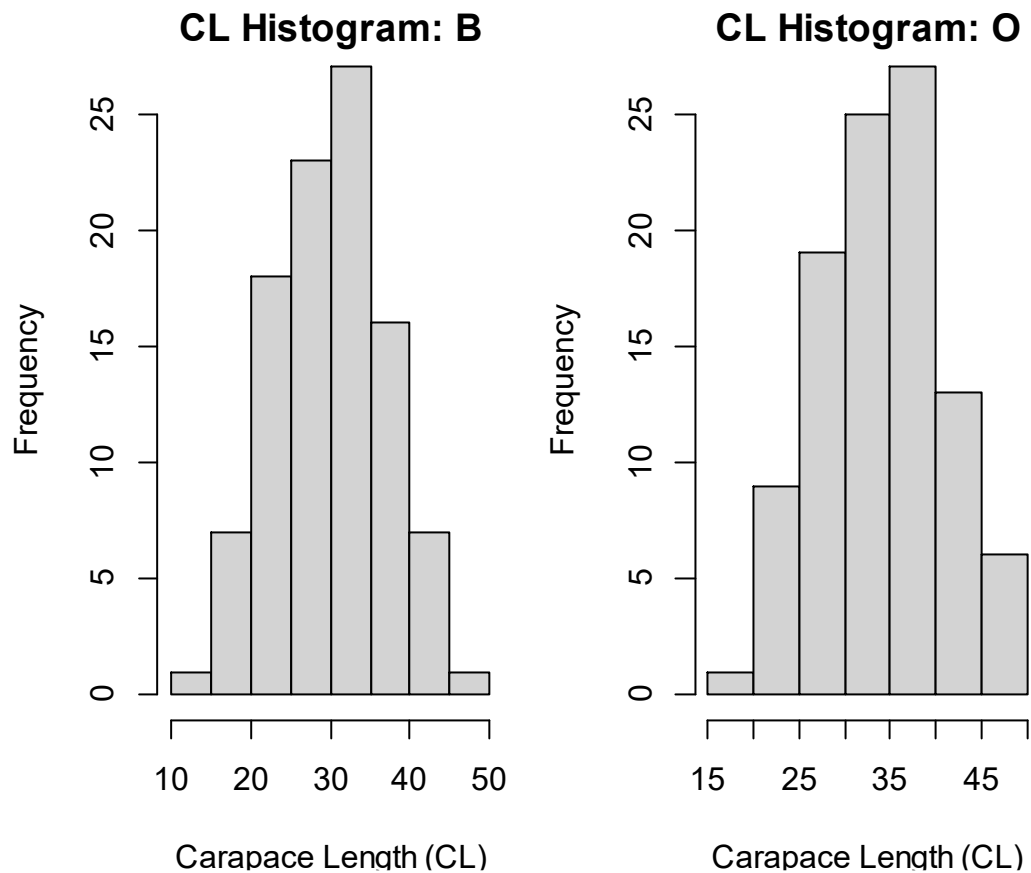
```
OUTLIERS <- boxplot(BD ~ interaction(sp, sex, sep = "_"), data =
crabs, plot = FALSE)
outliers_df <- data.frame(group = bp$names[bp$group], outlier_BD =
bp$out)
outliers_df
```

Q3.

a. Histograms of carapace length (CL) for each species

```
par(mfrow = c(1, 2))
hist(crabs$CL[crabs$sp == levels(crabs$sp)[1]],
     main = paste("CL Histogram:", levels(crabs$sp)[1]),
     xlab = "Carapace Length (CL)")

hist(crabs$CL[crabs$sp == levels(crabs$sp)[2]],
     main = paste("CL Histogram:", levels(crabs$sp)[2]),
     xlab = "Carapace Length (CL)")
par(mfrow = c(1, 2))
```



b.

```
by(crabs$CL, crabs$sp, summary)

SKEW <- function(x){
  x <- x[is.finite(x)]
  m <- mean(x); s <- sd(x)
  mean(((x - m)/s)^3)
}
by(crabs$CL, crabs$sp, SKEW)
```

```
> by(crabs$CL, crabs$sp, summary)
crabs$sp: B
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 14.70  24.85   30.10   30.06  34.60   47.10
-----
crabs$sp: O
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 16.70  29.20   34.55   34.15  39.40   47.60
```

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```
crabs$sp: B  
[1] 0.02571518  
-----  
-----
```

```
crabs$sp: O  
[1] -0.1536776
```

For specie B, it is positively skewed

For specie O, it is negatively skewed

Question 4

a.

```
MEAN <- colMeans(crabs[, c("FL", "RW", "CL", "CW", "BD")])  
MEAN
```

FL	RW	CL	CW	BD
15.5830	12.7385	32.1055	36.4145	14.0305

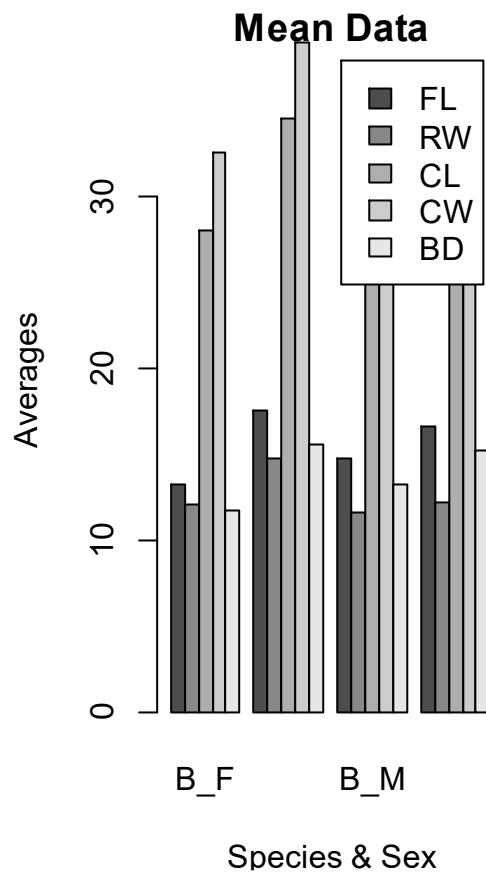
b.

```
mean <- aggregate(crabs[, c("FL", "RW", "CL", "CW", "BD")],  
                  by = list(sp = crabs$sp, sex =  
crabs$sex),  
                  FUN = mean)  
  
grp_labels <- paste(mean$sp, mean_by_group$sex, sep = "_")  
  
means_mat <- t(as.matrix(mean[, c("FL", "RW", "CL", "CW", "BD")]))  
colnames(means_mat) <- grp_labels  
  
barplot(means_mat, beside = TRUE, legend.text =  
rownames(means_mat),  
        xlab = "Species & Sex", ylab = "Averages",  
        main = "Mean Data")
```

c. For species difference, comparing Blue-Orange species within the same sex, CL shows the largest species difference whereas in sex variation, comparing male-female within each species, RW shows the smallest sex variation.

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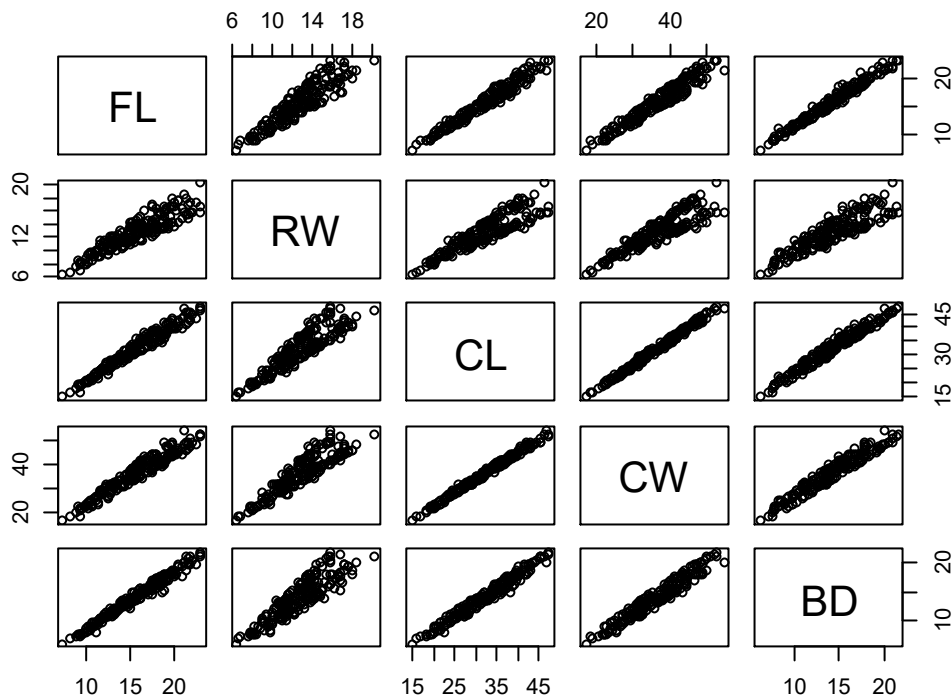
c.

Question 5

a. Scatter plot

```
pairs(crabs[, c("FL", "RW", "CL", "CW", "BD")],  
      main = "PAIRS PLOT")
```

PAIRS PLOT



b. From the pair plot above, we can see that, the variables all have a positive correlation. This means they all increase in proportion to one another.

Q6. THE REPORT

The “crabs” dataset, available in R mass package, is a data frame with 200 rows and 8 columns, describing 5 morphological measurements on 50 crabs each of two-colour forms and both sexes and of the species *Leptograpsus variegatus* collected at Fremantle, W. Australia. The dataset consists of 200 observations and 8 variables. It had its measurements from two colour forms (species types), which are B (Blue) and O (Orange) and Two sexes; F (Female) and M (Male). From the dataset, each combination of species and sex contains 50 crabs, making the dataset perfectly balanced.

The averages of each provide a crab morphology across all groups are, FL=15.5830, RW=15.5830 etc as seen in question 2. From question 3, the dataset is perfectly balanced, meaning that each of the four groups contains exactly 50 observations and no group is better than the other.

From the side-by-side boxplot of Body Depth (BD), we realized that male crabs show a higher median body depth than females. From the histograms that were plotted for Carapace Length (CL), Species B has mean= 30.06, median=30.10 and Skewness=0.026 which is slightly positively skewed and the rest as it can be seen in the boxplot

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From q6, all variables show strong positive correlations. This means as one measurement increases, others tend to increase as well.

In conclusion the dataset is perfectly balanced, ensuring fair statistical comparisons. Orange crabs are generally larger than blue crabs. Male crabs are larger than female crabs across all measurements. Body measurements are strongly positively correlated. Carapace length distributions are approximately normal with minimal skewness.